

# Config Parameter Cheat-Sheet

All simulation behaviour is driven by nested Python dicts. This sheet covers every parameter, its sensible range, its purpose, and which function consumes it.

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## 1. Disturbance Landscape (`disturbance_config`)

Used by: `DisturbanceLandscape.__init__()`

### **random**

| Parameter | Range               | Purpose                                                                                                            |
|-----------|---------------------|--------------------------------------------------------------------------------------------------------------------|
| seed      | any int, or<br>None | Seeds <code>np.random.seed()</code> . None → timestamp-based (non-reproducible). Set an int for reproducible runs. |

### **raster**

| Parameter | Range | Purpose                                                                                       |
|-----------|-------|-----------------------------------------------------------------------------------------------|
| band      | 1–N   | Which band of the input <code>.tif</code> to read as the patch label raster. Almost always 1. |

### **`types.type1` / `types.type2`**

Each type block has the same structure. The two types share all patches between them.

| Parameter                                          | Range                              | Purpose                                                                                                                                                                                                        | Used in                                        |
|----------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| proportion_of_patches                              | 0–1<br>(two types should sum to 1) | Fraction of SNIC patches randomly assigned to this type.                                                                                                                                                       | <code>_setup_types_and_assign_patches()</code> |
| name                                               | string                             | Human-readable label, used in plots.                                                                                                                                                                           | diagnostic plots                               |
| initial_cover.target_mean                          | 0.05–0.98                          | Mean of the Beta distribution for initial canopy cover (in 0–1 units, so 0.85 = 85%).                                                                                                                          | <code>_calculate_initial_forest_cover()</code> |
| initial_cover.kappa                                | 5–50                               | Concentration of the Beta distribution. Higher = tighter cluster around the mean.<br>kappa=20 with mean=0.85 gives roughly $\pm 10\%$ spread.<br>Internally:<br>alpha = mean * kappa, beta = (1–mean) * kappa. | <code>_calculate_initial_forest_cover()</code> |
| proportional_loss.mixture.catastrophic_probability | 0.001–0.2                          | Probability that a patch                                                                                                                                                                                       | <code>_generate_loss_rasters()</code>          |

| Parameter                                    | Range   | Purpose                                                                                                                                       | Used in                  |
|----------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
|                                              |         | draws its loss from the catastrophic component rather than the background component. E.g. 0.02 = 2% of patches experience severe disturbance. |                          |
| proportional_loss.mixture.background_alpha   | 0.1–2.0 | Alpha shape for the background Beta distribution. Low alpha + high beta concentrates draws near zero (most patches lose very little).         | _generate_loss_rasters() |
| proportional_loss.mixture.background_beta    | 3–20    | Beta shape for background loss. Higher = more mass near zero.                                                                                 | _generate_loss_rasters() |
| proportional_loss.mixture.catastrophic_alpha | 3–15    | Alpha shape for catastrophic Beta. High alpha + low beta concentrates draws near 1                                                            | _generate_loss_rasters() |

| Parameter                                   | Range   | Purpose                                      | Used in                  |
|---------------------------------------------|---------|----------------------------------------------|--------------------------|
|                                             |         | (near-total loss).                           |                          |
|                                             |         | Beta shape for                               |                          |
| proportional_loss.mixture.catastrophic_beta | 0.3–2.0 | catastrophic loss. Lower = more mass near 1. | _generate_loss_rasters() |

## thresholds

| Parameter           | Range | Purpose                                                                                                                                                     | Used in                                                                |
|---------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
|                     |       | A pixel is labelled disturbed = True when final_cover < this value AND cover_loss > 0. Also used as the default pivot for the interpreter definition curve. |                                                                        |
| final_cover_percent | 5–30  |                                                                                                                                                             | _update_disturbed_mask(), InterpreterField._compute_base_probability() |

## simulation

| Parameter   | Range | Purpose                                                                                                        | Used in                         |
|-------------|-------|----------------------------------------------------------------------------------------------------------------|---------------------------------|
| max_retries | 1–20  | How many times to re-draw the loss rasters if zero disturbed pixels result. Rare with sensible mixture params. | DisturbanceLandscape.__init__() |

## 2. Sensor (sensors[ ].config)

Used by: SensorField.\_run\_pipeline()

Type-specific keys under types.type1 / types.type2 override the top-level defaults for that type.

## signal\_transform

| Parameter  | Range    | Purpose                                                                                                                                                | Used in                                                 |
|------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| max_signal | 500–5000 | Maximum sensor signal value at 100% cover. Scales the whole signal range.                                                                              | _compute_signal_difference_step() via cover_to_signal() |
| curve_k    | 0.5–10   | Curvature of the log-asymptotic cover→signal transform. Higher = more concave (sensor saturates earlier). k=1 is nearly linear; k=5 saturates quickly. | _compute_signal_difference_step() via cover_to_signal() |

## gaussian\_patch\_noise

| Parameter               | Range  | Purpose                                                                                                                                             | Used in                                                        |
|-------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| std_proportion_of_range | 0–0.15 | Per-patch noise std as a fraction of the total signal range observed in the scene. 0.02 = noise std is 2% of the signal range. Set to 0 to disable. | _add_patch_noise_step() via add_gaussian_patch_noise_by_type() |

## lowpass\_filter

| Parameter   | Range                       | Purpose                                                                                                                                   | Used in                                          |
|-------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| kernel_size | 0–15 (odd integers typical) | Size of the 2D uniform smoothing kernel in pixels. 0 or 1 = no smoothing. Larger values blur patch boundaries and spread noise spatially. | _apply_lowpass_step() via apply_lowpass_filter() |

## **spatial\_noise**

| Parameter                     | Range         | Purpose                                                                                                                                     | Used in                                                                                  |
|-------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| spatial_autocorr_distance     | 2–50 (pixels) | Correlation length of the gstools Gaussian random field added as spatially structured noise. Larger = broader correlated patches of noise.  | <code>_add_spatial_noise_step()</code> via <code>add_spatially_correlated_noise()</code> |
| amplitude_proportion_of_range | 0–0.3         | Amplitude of spatial noise as a fraction of the signal range. <code>0.1</code> = $\pm 10\%$ of its range. Set to <code>0</code> to disable. | <code>_add_spatial_noise_step()</code> via <code>add_spatially_correlated_noise()</code> |

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## **3. Interpreter Field (interpreter.config)**

Used by: `InterpreterField._run_pipeline()`

Per-type overrides go in `types.type1 / types.type2` using the same key names. Any key omitted from the type block falls back to the top-level default.

### **detection\_curve**

Controls how strongly the interpreter perceives disturbance as a function of **cover loss**.

| Parameter  | Range               | Purpose                                                                                                                  | Used in                                  |
|------------|---------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| pivot_loss | 5–40 (% cover loss) | Cover loss value where the interpreter has 50% probability of detecting disturbance. Lower = more sensitive interpreter. | <code>_compute_base_probability()</code> |
| slope      | 0.05–1.0            | Steepness of the detection logistic. Higher = sharper transition from "not detected" to "detected".                      | <code>_compute_base_probability()</code> |

## definition\_curve

Controls how the interpreter applies the forest/non-forest **threshold** to final cover.

| Parameter                    | Range                                                                                                                                                                                                                               | Purpose                                                                      | Used in                                  |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------|
| slope                        | 0.05–1.0                                                                                                                                                                                                                            | Steepness of the definition logistic around the threshold. Higher = sharper. | <code>_compute_base_probability()</code> |
| threshold_override (or None) | If set, overrides the <code>final_cover_percent</code> % cover from the disturbance config as the interpreter's threshold. None = use the truth threshold. Useful to simulate interpreter/map disagreement on the threshold itself. |                                                                              | <code>_compute_base_probability()</code> |

## gaussian\_patch\_noise

| Parameter                           | Range | Purpose                                                                                                                                                                                 | Used in                              |
|-------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| std_dev_percent (percentage points) | 0–20  | Per-patch Gaussian noise on the probability value. 8.0 means noise std = 8 percentage points. Added as an offset to the base probability before clipping to [0,1]. Set to 0 to disable. | <code>_apply_patch_gaussian()</code> |

## beta\_patch\_uncertainty

Draws a per-patch scaler from Beta(alpha, beta) and uses it to shrink the probability toward 0.5:  $p_{\text{new}} = p + \text{scaler} \times (0.5 - p)$ .

| Parameter | Range  | Purpose                                                                                                                                    | Used in                          |
|-----------|--------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| alpha     | 0.1–10 | Alpha shape of the Beta distribution for the shrink scaler. Low alpha + high beta = most patches barely shrink toward 0.5 (scaler near 0). | <code>_apply_patch_beta()</code> |
| beta      | 0.1–20 | Beta shape of the shrink scaler distribution. Higher beta relative to alpha = stronger pull toward 0.5 in most patches.                    | <code>_apply_patch_beta()</code> |

## lowpass\_filter

| Parameter   | Range | Purpose                                                                                                                           | Used in                                                                    |
|-------------|-------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| kernel_size | 0–15  | Spatial smoothing of the probability field. Simulates edge effects and spatial coherence in interpreter perception. 0 = disabled. | <code>_apply_lowpass_step()</code> via <code>apply_lowpass_filter()</code> |

## spatial\_uncertainty

Generates a spatially correlated field of scalars in  $[0, \text{max\_scaler}]$  and uses them to shrink probabilities toward 0.5 across the landscape.

| Parameter                 | Range         | Purpose                                                                                                                         | Used in                                                                                       |
|---------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| spatial_autocorr_distance | 2–50 (pixels) | Correlation length of the spatial uncertainty field. Larger = broader regions of high/low interpreter uncertainty.              | <code>_apply_spatial_uncertainty()</code> via <code>spatial_uncertainty_scaler_field()</code> |
| max_scaler                | 0–0.8         | Maximum shrink-toward-0.5 scaler. 0.3 means uncertainty can pull probabilities at most 30% of the way toward 0.5. 0 = disabled. | <code>_apply_spatial_uncertainty()</code> via <code>spatial_uncertainty_scaler_field()</code> |

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## 4. Interpreter Agent (InterpreterAgent config)

Used by: `InterpreterAgent.__post_init__()` and `InterpreterAgent.transform_probabilities()`

This controls **individual human label noise** applied at sampling time (not the full raster).



| Parameter           | Range                       | Purpose                                                                                                                                                                                                            | Used in                   |
|---------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| interpreter_id      | string                      | Unique name for this agent.                                                                                                                                                                                        | identification            |
| consistency_std_pct | 0–20<br>(percentage points) | Per-sample Gaussian noise. Drawn fresh for each label. Models within-interpreter inconsistency.                                                                                                                    | transform_probabilities() |
| bias_std_pct        | 0–15<br>(percentage points) | Directional bias drawn once per agent from $N(0, \text{bias\_std\_pct})$ . Positive = systematically over-labels disturbance; negative = under-labels.                                                             | __post_init__()           |
| confidence_std      | –1 to +1                    | Directly sets the confidence scalar (not drawn stochastically — see code note). Positive = pushes all probabilities toward 0 or 1 (overconfident). Negative = pulls all probabilities toward 0.5 (underconfident). | transform_probabilities() |
| seed                | int or None                 | Seeds this agent's RNG. None = timestamp + hash of interpreter_id.                                                                                                                                                 | __post_init__()           |

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## 5. Collection / Experiment Config (experiment\_cfg)

Used by: LandscapeStackCollection.from\_config()

| Parameter                    | Range       | Purpose                                                                                                             |
|------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------|
| n_stacks                     | 1–200+      | Number of landscape stacks to build. More = better Monte Carlo coverage but slower.                                 |
| patch_dir                    | path string | Directory containing SNIC .tif patch rasters. One raster is assigned to each stack.                                 |
| patch_selection_seed         | int or None | Seeds the random selection of rasters from patch_dir. Fix for reproducible stack assignments.                       |
| disturbance_seed_base        | int or None | Base random seed for DisturbanceLandscape. Stack $i$ gets seed = base + $i$ (if use_separate_seeds_per_stack=True). |
| use_separate_seeds_per_stack | bool        | If True, each stack gets a unique seed (base + $i$ ). If False, all stacks share the same seed (not recommended).   |

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## 6. Binary Classifier Parameters

Used by: `LandscapeStackCollection.buildBinaryClassifier()`

| Parameter                        | Range                    | Purpose                                                                                                                                                                                  |
|----------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>n_strata</code>            | 3–10                     | Number of equal-width strata on sensor values for training sample stratification.                                                                                                        |
| <code>samples_per_stratum</code> | 20–500                   | Samples drawn per stratum per stack. More = more stable threshold estimate.                                                                                                              |
| <code>sampling_seed</code>       | int or None              | Seeds the stratified sample draw.                                                                                                                                                        |
| <code>fp_to_fn_ratio</code>      | 0.1–10                   | Tradeoff weight: $\text{cost} = \text{fp\_to\_fn\_ratio} \times \text{FPR} + \text{FNR}$ . $>1$ penalises false positives more; $<1$ penalises false negatives more. $1.0$ = equal cost. |
| <code>interpreter_agent</code>   | InterpreterAgent or None | If provided, perturbs interpreter probabilities before thresholding to binary labels.                                                                                                    |

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## 7. Continuous Model Parameters

Used by: `LandscapeStackCollection.buildModel()` and `buildInterpreterProbModel()`

| Parameter                        | Range                    | Purpose                                                                                                                                                        |
|----------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>n_strata</code>            | 3–10                     | Strata for training sample stratification (on sensor values for <code>buildModel</code> ; on model probabilities for <code>buildInterpreterProbModel</code> ). |
| <code>samples_per_stratum</code> | 20–500                   | Samples per stratum per stack.                                                                                                                                 |
| <code>sampling_seed</code>       | int or None              | Seeds the sample draw.                                                                                                                                         |
| <code>interpreter_agent</code>   | InterpreterAgent or None | If provided, transforms base interpreter probabilities before fitting the Richards curve.                                                                      |

### Richards curve (fitted, not configured directly)

The model fits  $p(x) = (1 + \exp(-b \times (x - x_0)))^{-nu}$ . The fitted parameters are stored in the model dict for inspection:

| Parameter | Meaning                                                        |
|-----------|----------------------------------------------------------------|
| x0        | Inflection point — sensor value at which probability is ~0.5   |
| b         | Slope / steepness                                              |
| nu        | Shape / asymmetry. nu=1 = standard logistic; nu>1 = asymmetric |

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## 8. Spatial Monte Carlo Area Estimation

Used by: `LandscapeStackCollection.calc_area_distribution_spatial_threshold_for_stack()`

| Parameter | Range                     | Purpose                                                                                                                                   |
|-----------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| n_iter    | 100–5000                  | Number of Monte Carlo threshold surface draws. More = smoother area distribution.                                                         |
| m         | 0–1, or (low, high) tuple | Mean of the spatially correlated threshold surface. Can be a fixed float or a (low, high) tuple to draw uniformly each iteration.         |
| l         | 2–100 pixels, or tuple    | Spatial correlation length of the threshold surface. Larger = broader patches of high/low threshold. Can be fixed or drawn per iteration. |
| sigma     | 0.05–0.5                  | Std dev of threshold surface about m. Larger = more spatially variable threshold, wider area distribution.                                |